

Fatima ' s Backend Development Roadmap

JavaScript + Node.js + Elasticsearch (9-Week Plan)

Phase 1: JavaScript Foundation (Weeks 1 – 3)

Goal: Build a strong base in core JavaScript concepts — callbacks, async/await, promises, closures, hoisting, and event loop. Yeh stage sabse important hai kyunki Node.js isi language pe run hota hai.

- 1 Variables, Data Types, Operators
- 2 Functions, Arrow Functions, Scope & Hoisting
- 3 Arrays & Objects Manipulation
- 4 Callbacks, Promises, Async/Await (yeh concept async ka hota hai jo code ko delay ke bina run karta hai)
- 5 Event Loop & Call Stack
- 6 Closures (function ke andar function access to outer variable)
- 7 DOM Manipulation & Events
- 8 Fetch API (for calling external APIs)

Mini Projects: Console-based To-Do App, Weather API App

Phase 2: Node.js Mastery (Weeks 4 – 7)

Learn server-side JavaScript using Node.js and Express. Yeh part tumhe real backend developer banayega.

- 1 Node.js Fundamentals (event loop, non-blocking I/O, modules)**
- 2 File System, NPM packages, Environment Variables**
- 3 Express.js Framework, Routes, Middleware, CRUD APIs**
- 4 Error Handling, Async Middleware, API Structure**
- 5 JWT Authentication, Password Hashing using bcrypt**
- 6 Testing APIs with Postman**

Mini Projects: User CRUD API, Auth System with JWT

Phase 3: Elasticsearch Integration (Weeks 8 – 9)

Integrate Elasticsearch with Node.js to build powerful search APIs. Tum already Elasticsearch use karti ho — ab usse Node.js ke sath connect karna seekho taake full project ready ho.

- 1 Installing & Running Elasticsearch Cluster
- 2 Elasticsearch Node.js Client Setup
- 3 Indexing Data and Searching via API
- 4 Filters, Pagination, Sorting
- 5 Error Handling and Performance Optimization

Mini Project: Real Estate Search API (index + search properties)

A Note for Fatima

Fatima, remember this — learning tech is not a race, it's a journey. Tumne already ek strong step liya hai apni career direction clear karne ka. Stay consistent, 2 hours daily ka plan follow karo, weekends par projects banao, aur har week ek concept master karo. Within 3 months, tumhara backend level strong ho jayega. Keep going — the future you will thank you